Tutorial 3 –Experimental error

Background

- When adding and subtracting, the answer should have the same number of decimal places as the limiting term. The limiting term is the number with the least decimal places.
- When multiplying/dividing, the answer should have the same number of significant figures as the limiting term.
- Always retain more digits than necessary during the calculation and round off to the appropriate number of digits in the end.
- Usually, just one significant figure is kept in the error (Never mind it is absolute error or relative error).
- Don't forget that the significant figure of magnitude should be modified by the mantissa of uncertainty for the final result.

Exercises

T3-1. Calculators are great devices. Their invention has allowed for quick computation where manipulation of numbers needs to be done rapidly and accurately. But they are only as good as the numbers put into them. The calculator cannot determine how accurate each set of numbers is, and the answer given must be assessed by the user for reliability.

Consider two separate mass measurements: 0.75 g and 0.241 g, when these masses are added together, the result on a calculator is 0.75 g + 0.241 g = 0.991 g. How should you report the total mass as the answer properly? How many significant figures in your result?

T3-2. Check if the significant figures in the final answer are correct. If they are wrong, please correct them.

$$Log\ (1.237 \times 10^{-3}) = -2.908$$

$$10^{2.384} = 242.169$$

T3-3. a) Calculate the answer to the following equation and give your answer with the correct significant figures.

$$\frac{2.1 \times (1.45 + 0.984)}{8.115 \times 1.70} - 6.974$$

b) Calculate the absolute and relative uncertainties of the answer to the following equation. Express the answer and the uncertainties with the correct significant figures.

$$6.843\ (\pm\ 0.008) \times 10^4 \div [2.09\ (\pm\ 0.04 - 1.63\ (\pm\ 0.01)]$$

T3-4. Write down the answer and uncertainty with the correct significant figures.

a. $\frac{0.002364(\pm 0.000\ 003)}{0.02500(\pm 0.000\ 05)} =$

b. If the mass is 4.635(±0.002) g and the volume is 8.12(±0.05) mL. How much the density?

c. To prepare a solution, 2.634 ± (0.002) g of NaCl is weighed and dissolved it in a volumetric flask with a volume of 100.00 (± 0.08) mL. The molar mass of NaCl is 58.442(±0.006) g/mol. How much the concentration of NaCl solution in mol/L?